

VN1K is a pangenome-informed multi-omics and phenomics resource for the Vietnamese population

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The population of Vietnam remains underrepresented in global genomic databases. Here, we present VN1K, a resource of multi-omics and phenotypic information for 1011 unrelated Vietnamese individuals. We present high-depth short-read whole-genome sequencing data for all samples along with various -omics datasets. Using a high-sensitivity variant detection pipeline, which includes a pangenome graph reference and a deep-learning framework, we identify approximately 42 million variants with 7 million short insertions/deletions and 90 thousand structural variants. VN1K also features a whole-genome methylation profile based on long read sequencing. We create a genotype imputation panel with high accuracy on the Vietnamese population, allowing us to identify variants with significantly different allele frequencies in the Vietnamese population compared to other populations. We establish the functional relevance of some of these variants, particularly those in genes associated with genetic disorders, immune diseases, and drug responses, by integrating the allele frequency differences with known genotype–phenotype associations and clinical annotations. Further, we map various loci related to hepatitis B virus infection, triglyceride levels, LDL-C levels, serum glucose levels, HbA1c levels, and levels of two liver enzymes (ALT and AST). The VN1K dataset is accessible via genome.vinbigdata.org, an integrated platform with both linear and graph-based genome browsers.

Vietnam is profoundly underrepresented in the current global genomic databases despite the large population and extensive genetic and phenotypic diversity. As of December 2024, Vietnam has a population of ~100 million people and is ranked first in the mainland region of Southeast Asia and the 16th worldwide with a long and complex migratory history, resulting in extensive ethnolinguistic and genetic diversity¹. As Vietnam is located in a key geopolitical region connecting South, East, and Southeast Asia, understanding its genetic structure is critical to further elucidate the genetic diversity of the larger neighboring areas. Despite these facts, the current widely-used whole genome sequencing (WGS) dataset for Vietnamese comprises only 99 samples of Kinh in Ho Chi Minh City, Vietnam (KHV) in the 1000 Genomes Project (1kGP)², severely under-representing the

approximately 100 million people across the 64 provinces. Recent WGS projects for the Vietnamese population were performed with just either a few samples with high coverage (a trio at 30X)³ or few hundreds of samples with low-to-medium coverage (101 WGS at 4x from 1kGP-KHV, 105 WGS at 15x and 200 WGS from in-house data)⁴. In recent Asian genome projects, including the GenomeAsia100K Pilot Project (GAsP)⁵, in which Southeast Asian samples account for nearly 20%, the genetic data from the Vietnamese population is not present. This lack of knowledge highlights the underrepresentation of Vietnamese population in the global genomics, possibly due to a long-term exclusion from Vietnamese origins from the database.

Despite the remarkable expansion of genome projects, current studies have mostly focused on European individuals (or Caucasian

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ancestry), which capture around 80% of participants in genome-wide association studies (GWAS)⁶. Asians, accounting for 60% worldwide population with diverse ethnicities (20007900), are still under-represented in large databases such as gnomAD⁷ or TOPMed⁸. Although some groups such as South Asians and East Asians have been thoroughly studied in 1kGP, many of them are often lumped together in large projects. Currently, an increasing number of Asian genome projects have been conducted, leading to the robust discovery of unreported variants compared to European-centric genomic datasets^{9–12}. Notably, Southeast Asian genomes such as Malaysian in SG10K yield more unreported variants than both South and East Asian¹². This indicates that Southeast Asia, which accounts for 8.5% of the world's population and has a long and complex history of human migration within the region and from South and East Asia regions, is greatly understudied and requires more GWAS studies. Inclusion of the Southeast Asian genomes in the genomic databases also significantly reduced the error rates of genomic imputation for ethnically related populations such as Melanesians and Papuans from Oceania. Further investigation of the genetics of Southeast Asians, including Vietnamese, will help uncover rare and population-specific variants, inform variant prevalence, and provide a comprehensive reference genome.

Here we present VNIK, a large-scale genome program for the Vietnamese population (1011 individuals) with high-depth WGS (37X median, 30X for 98% samples) coupled with latest genomic sequencing technologies and functional assays (PacBio and Oxford Nanopore Technologies, Axiom™ Precision Medicine Diversity Array, and RNA sequencing) along with a wide-range of common phenotypic traits

($n = 34$ in total, including demographic, anthropometric, biochemical, and clinical traits). As a result, we identify nearly 8.5 million variants that are possibly specific to the Vietnamese population, establishing both a detailed map of Vietnam's genetic diversity and a high-accuracy imputation reference. A pangenome graph from long- and short-read data is constructed to improve structural variant detection, and unique patterns in pharmacogenes and immunogenes are characterized. Comprehensive variant annotation is performed to highlight pathogenic alleles of clinical relevance. In particular, genome-wide methylation profiles are reported to refine functional interpretation of variants. Genome-wide association analyses across 12 phenotypes confirm VNIK's utility despite limited sample size, and population analyses reveal regional and global genetic distinctions. Overall, our study presents a comprehensive dataset, stringently curated and functionally annotated to investigate their phenotypic and potential clinical impacts in the Vietnamese and related populations of Asian-origin, and contributes to research and applications in the field across Southeast Asia and the world.

Results

VNIK reveals a large number of genomic variants and improves imputation accuracy

VNIK comprises a large collection of 1011 Vietnamese individuals geographically representative for 53 of the total 63 provinces and cities of Vietnam (Fig. 1a). Our stringent pipeline for short-read WGS data defined a high-quality variant callset with 34,875,006 bi-allelic Single Nucleotide Polymorphisms (SNPs), 193,553 multi-allelic SNPs, 5,111,833 bi-allelic short insertion/deletions (INDELs), and 1,433,546 multiallelic

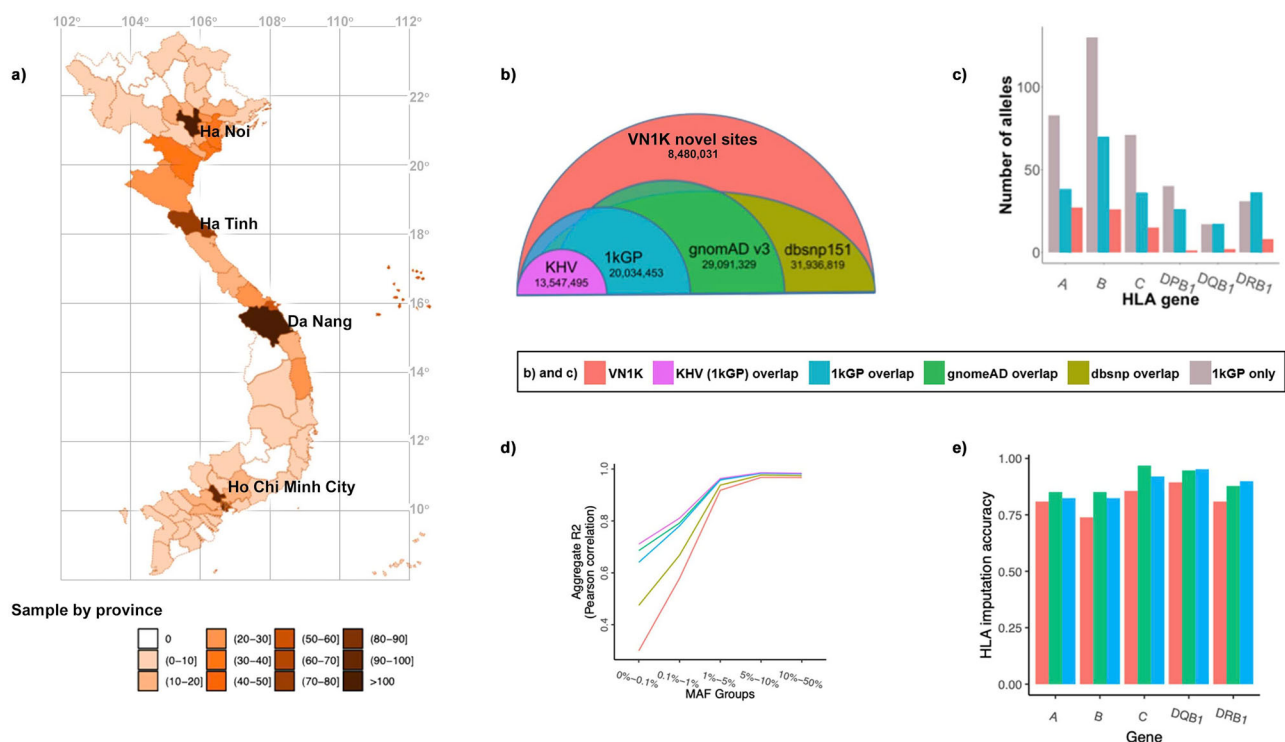


Fig. 1 | Sample collection, statistical analysis of the VNIK-specific sites, and the application of SNP-HLA imputation techniques to the VNIK dataset. The 1011 unrelated Vietnamese individuals across 53/63 provinces and cities revealed millions of variants not reported in existing databases. **a** Geography of collected samples 1011 Vietnamese individuals declared as Kinh-ethnic for at least 3 generations and confirmed as healthy by their general health check information. Participants came from 53 of the total 63 administrative divisions of Vietnam (the colored areas); the provinces/cities shown in the map indicated the main collection site. **b** The total number of variants in VNIK overlaps with 1kGP-HC, gnomAD v3 and

dbsnp151. **c** The number of HLA alleles class I and II in the 1kGP-HC and the VNIK-specific alleles. **d** Imputation accuracy measured by aggregated R2 of all imputed variants in different MAF bins for Vietnamese-Kinh population based on 3 single panels (1kGP-HC, SG10K, VNIK) and 2 merge panels (VNIK_1kGP-HC and VNIK_SG10K). **e** HLA imputation accuracy comparison between different imputation panels. Abbreviations: gnomAD: The Genome Aggregation Database; dbsnp151: Single Nucleotide Polymorphism Database version 151; 1kGP: 1000 Genomes Project; 1kGP-HC: 1000 Genomes Project, High Coverage Dataset; SG10K: Singaporean Genomes Project.

complex sites. After a process of multi-allelic to bi-allelic normalization, the callset contained 35,073,660 SNPs and 7,094,874 INDELs that sum up to 42,168,534. Furthermore, a large number of Copy Number Variants (CNVs) were identified with a total of 21,255, consisting of 17,192 deletions (DELs) and 4063 duplications (DUPs). Compared to 1000 Genomes Project high coverage (1kGP-HC), gnomAD v3 and dbSNP151 (Database of Single Nucleotide Polymorphisms, see the Data Availability section), we detected 8,480,031 variants likely specific for the Vietnamese and possibly other under-represented populations of Asian-origin as well (Fig. 1b). Many of these Vietnamese-specific variants are in important functional regions. For instance, we found 166 human leukocyte antigen (HLA) class I and II alleles, of which 79 were possibly specific to the Vietnamese population (Fig. 1c). More details of samples and variants can be found in Supplementary Figs. 1–3 and Supplementary Data 1a–i.

To further investigate the VN1K variants, we compared them to the most recent Vietnamese genomic database, which included a collection of 406 individuals, comprising 105 whole genomes, 200 whole exomes, and 101 published Vietnamese whole genomes from 1kGP (1kGP-KHV). Strikingly, our results showed that the vast majority (95.75%) of the VN1K variants are not in this database. This significant addition was not only derived from a larger number of 1011 samples, but more importantly, resulted from well-curated, high-depth sequencing (30–60x), and multi-platform sequencing data, including short-read data from Illumina and long-read data from PacBio and Oxford Nanopore Technologies (ONT) platforms.

The large number of Vietnamese-specific variants enabled more accurate imputation for the Vietnamese population than any other reference panels. In particular, the merged panel VN1K_SG10K has the best performance in imputing the genotype of the Vietnamese samples, which was validated using Affymetrix PMRA microarray data of 96 randomly selected samples. VN1K also led to the highest aggregate R^2 (i.e., the coefficient of determination) between the imputed and true genotypes across all bins of minor allele frequency (MAF) (Fig. 1d, Supplementary Table 1). Our imputation panel has remarkable translation implications, for example, in HLA imputation where the combined panel VN1K_1kGP-HC and VN1K provided comparable accuracy for both HLA class I and II genes which outperformed 1kGP-HC alone (Fig. 1e). Therefore, we expect that this Vietnamese-specific imputation panel could greatly benefit GWAS case-control studies on the Vietnamese population using SNP arrays. Indeed, we have recently assessed the application of population-specific imputation on estimating polygenic risk scores on the Vietnamese population and observed some improved performance¹³.

Pangenome graph reference combined with a deep-learning framework improves structural variant detection

To produce an extensive and stringent collection of SVs for VN1K, we combined and cross-validated SVs from short-read WGS data of all 1011 samples with long-read WGS data of 42 selected samples using PacBio and ONT platforms. To best represent and enable broad usage of such a comprehensive variant resource, we built a pangenome graph reference, hereby referred to as Vietnamese Pangenome Reference (VPR), using GRCh38 and the recently released T2T-CHM13 assembly¹⁴ as backbone. VPR integrated into graph the assemblies of two VN1K samples (one male and one female), which were deeply sequenced at 30X using PacBio and ONT platforms. These assemblies covered up to a total length of 3 Gbp (see more details in Supplementary Data 2). To improve the comprehensiveness and cross-database compatibility with public resources, we also incorporated into VPR assemblies of two KHV samples from HPRC dataset (Human Pangenome Reference Consortium)¹⁵ as additional tracks. Combined with a deep-learning framework, which utilized the Convolutional Neural Network model to refine the initial result from existing tools for SV calling (see the Methods section), VPR enabled accurate detection

of Vietnamese-specific SVs, which were difficult to obtain using just the linear reference GRCh38. In our study, Vietnamese-specific SVs are defined as variants that are common in the Vietnamese population ($AF \geq 5\%$ in VN1K) but rare or absent in other populations ($AF < 1\%$).

Overall, our pipeline detected 5046 ± 285 , 2848 ± 362 , 666 ± 74 , and 61 ± 11 (mean $\pm$ SD) number of SVs per individual for DEL, INS, DUP, and ITX (intra-chromosomal translocations), respectively (Fig. 2a, b). Our further analysis on the total of 88,241 SVs merging across all VN1K samples showed that rate of overlapping (short-read vs. long-read) between DEL was lower than that of INS in VN1K (Fig. 2c) and rate of overlapping SVs between VN1K and 1kGP-EAS (East Asian) was much higher than that of 1kGP-non-EAS, as expected (Fig. 2d). By examining the SVs that were not presented in 1kGP-HC (represented by the blue and orange areas in Fig. 2d), we found a significant SV subset with high AF, which potentially represented the SVs specific to the Vietnamese population (Supplementary Data 3a–b). Furthermore, we found that VPR significantly improved the SV calling compared to GRCh38, for both short-read and long-read data. As exemplified in a VN1K sample, there were 9343 and 7816 DELs detected with VPR compared to 8774 and 3762 DELs with GRCh38 using long-read and short-read data, respectively (Supplementary Fig. 4). For insertions, short-read discovery increased substantially (2296 with GRCh38 to 8769 with VPR), whereas long-read insertion counts were broadly comparable between the two (11,877 with GRCh38 versus 11,163 with VPR). Our further analyses on two additional KHV samples showed that VPR preserved a large number of SVs detected with GRCh38 while additionally recovering SVs that are missed by GRCh38, for both short-read and long-read data (Supplementary Figs. 5–10). The ability to capture the diverse and population-specific SVs through a graph-based pangenome reference and deep learning framework could unlock new insights into the genomic landscape of the Vietnamese population.

Functional genomic annotation analysis and implications in genetic disease screening

To investigate rare mutations, doubletons, and singletons that are unique to the Vietnamese population, we categorized the SNPs, INDELs, and SVs in VN1K by AF and compared them to current popular databases, including 1kGP-HC, gnomAD v3, and CCDG (Centers for Common Disease Genomics, see the Data Availability section) (Fig. 3a). Of all variants that are not in such databases, intronic and intergenic variant regions accounted for 54.3% and 29.6%, respectively. There were only 0.95% of variants in the coding and splicing regions (Fig. 3b). We defined an important SV as the one that overlaps with multiple regions of a gene, the region at the higher level of the functional hierarchy (in the order of CDS/exon, UTR, intron, flanking). Two ranges of flanking length were specified by default (i.e., 5Kb and 50Kb). The SVs in intronic and exonic regions attributed to 59.2% and 5.29%, respectively (Fig. 3c). Importantly, we found that nearly half of the VN1K-specific single nucleotide variants (SNV) were missense, 22.06% were splicing, and 21.72% were synonymous (Supplementary Fig. 11). The VN1K database enabled the discovery of specific variants in disease-associated genes such as *GCNT2* and *PMFBP1*, which exhibited a high prevalence (14.84%) in the Vietnamese population but rare (less than 0.5%) in other large populations of European. In contrast, the pathogenic variants associated with *HNRNPUL2* and *BSCL2*, which linked to congenital generalized lipodystrophy and breast carcinoma, were less prevalent in either the Vietnamese and East Asian (EAS) populations (less than 10% and 11.9%, respectively), yet more common in other populations (e.g., 28.93% in Europe and 28.12% in South Asia in 1kGP-HC). Notably, *PMFBP1*, *MTSS2*, *C9*, *SRDSA2*, and *HFE*, which were not featured in the American College of Medical Genetics and Genomics (ACMG) prenatal screening for genetic disorders, harbored known pathogenic variants with considerable AF (ranging from 1.48% to 3.26%) in the Vietnamese population (Supplementary Fig. 12). More

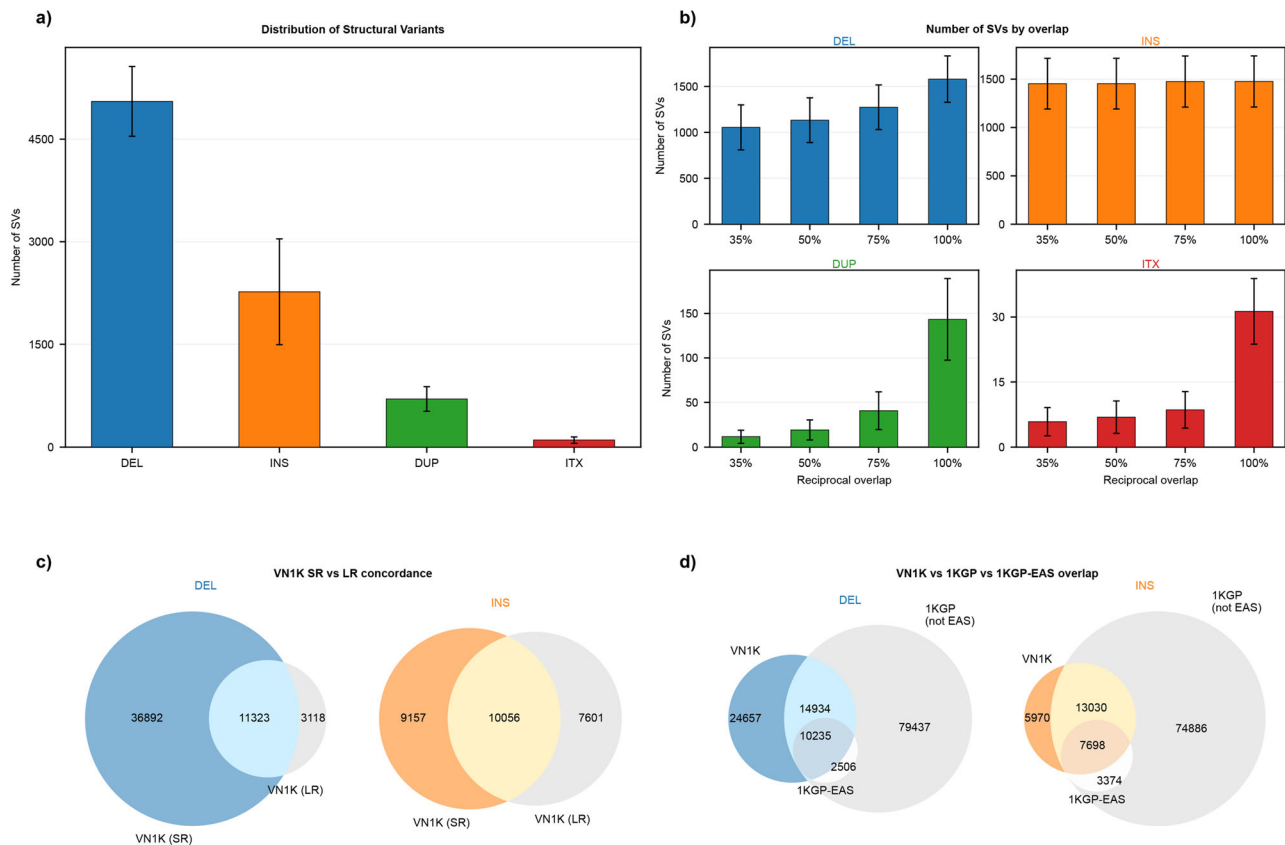


Fig. 2 | Structural variant analysis. The SV profiles demonstrate that long-read sequencing technologies significantly improve genomic coverage for the Vietnamese population, mirroring SV profile trends observed in other the 1kGP-HC populations. **a** Genomic distribution of detected SVs across different classes. The average number of detected variants per individual was 5046 ± 285 for DEL, 2848 ± 362 for INS, 666 ± 74 for DUP, and 61 ± 11 for ITX (all values represented as mean $\pm$ SD); **b** Reciprocal overlap thresholds (100%, 75%, 50%, and 35%) required between SV types; note that the numbers for INS is very similar across thresholds as expected; **c** Comparison of SV sites detected within VN1K using short-read

sequencing (Illumina) versus long-read sequencing (PacBio) platforms. The short-read sequencing data are from all 1,011 VN1K samples, while the long-read sequencing data were generated from 40 selected VN1K samples by a pooled design (10 samples per pool); **d** Comparison of SV sites between the VN1K and the 1kGP-HC datasets. Combining the VN1K short-read and long-read data revealed 24,657 SV-DELS and 5,970 SV-INSs that were previously undetected by 1kGP-HC. Abbreviations: SR: Short Reads; LR: Long Reads; EAS: East Asian; SV: Structural Variant; ITX: Intra-chromosomal Translocation; DUP: Duplication; INS: Insertion; DEL: Deletion; 1kGP-HC: 1000 Genomes Project High Coverage.

details of such functional impact information can be found in Supplementary Data 4a-d.

To explore more about the functional impact of VN1K, we performed a comprehensive data analysis to identify mobile element insertions (MEIs) using short-read data and cross-validated the results against long-read data. Our results showed that, on average, each sample has 1369 *ALU*, 198 *LINE1*, and 78 *SVA*. We observed substantial concordance between the two sequencing methods: 1144 and 1165 MEIs overlapped between long- and short-read calls, respectively, for two VN1K samples VN0007 and VN0920. While comparing with GENCODE MEI database from 5,675 genomes¹⁶, we found that the MEIs were mostly in the protein coding regions, with ~50% in introns and 35% outside any gene ("not gene" region) (see more details in Supplementary Figs. 13–16, Supplementary Table 2, Supplementary Data 5a-d).

The immunogenomics and pharmacogenomics references for the Vietnamese population

The VN1K genomic variant data also allows us to assess the genetics of the Vietnamese immune system and their potential implications in drug responses, an important contribution towards precision medicine in an under-represented country. The interplay between HLA and killer cell immunoglobulin-like receptors (KIR) was known to have significant impacts on immune function and disease susceptibility^{17–19}. First, to assess the generalizability of our data, we compared AF in

different ethnicities, and found that the HLA's AF exhibited consistent patterns with most public datasets of the Kinh ethnicity (e.g., KHV individuals in 1kGP-HC) but varied significantly across different ethnicities in Vietnam (e.g., Muong in Hoabinh province from Allele Frequency Net Database²⁰) (Fig. 4a, see more details in Supplementary Data 6a). *HLA-B*15:02*, which is highly associated with many Adverse Drug Reactions (ADR) such as carbamazepine-induced SJS/TEN, was the most common *HLA-B* allele in our cohort with a significantly higher AF than those observed in other populations of 1kGP-HC. In particular, *HLA-B*15:02* accounted for 12.07% of *HLA-B* allelic diversity in Vietnamese of VN1K (23.24% of individuals carried at least one copy of *HLA-B*15:02*) compared to 6.17% in East Asian (EAS) and 8.38% in South Asian (SAS) of 1kGP-HC. Allele and gene frequencies of KIR genes showed that all four framework genes (i.e., *KIR3DL3*, *KIR2DL1*, *KIR2DL4*, and *KIR2DL2*) and one telomeric gene (i.e., *KIR3DL1*) were most frequently present in on gene-content haplotypes of the Vietnamese population (Fig. 4b, see more details in Supplementary Data 6b).

Another clinically important and complex genomic region is the chromosome location that contains pharmacogenes (PGx), which affect the drug properties (i.e., pharmacokinetics and pharmacodynamics)²¹. VN1K included a total of 27,914 variants on the PGx genes, which were further assigned to 127-star alleles (or gene-level haplotype patterns that related to drug metabolisms) (Fig. 4c). For instance, the *CYP2D6*10* allele has an AF of 23.16%, suggesting a key genetic contribution to the

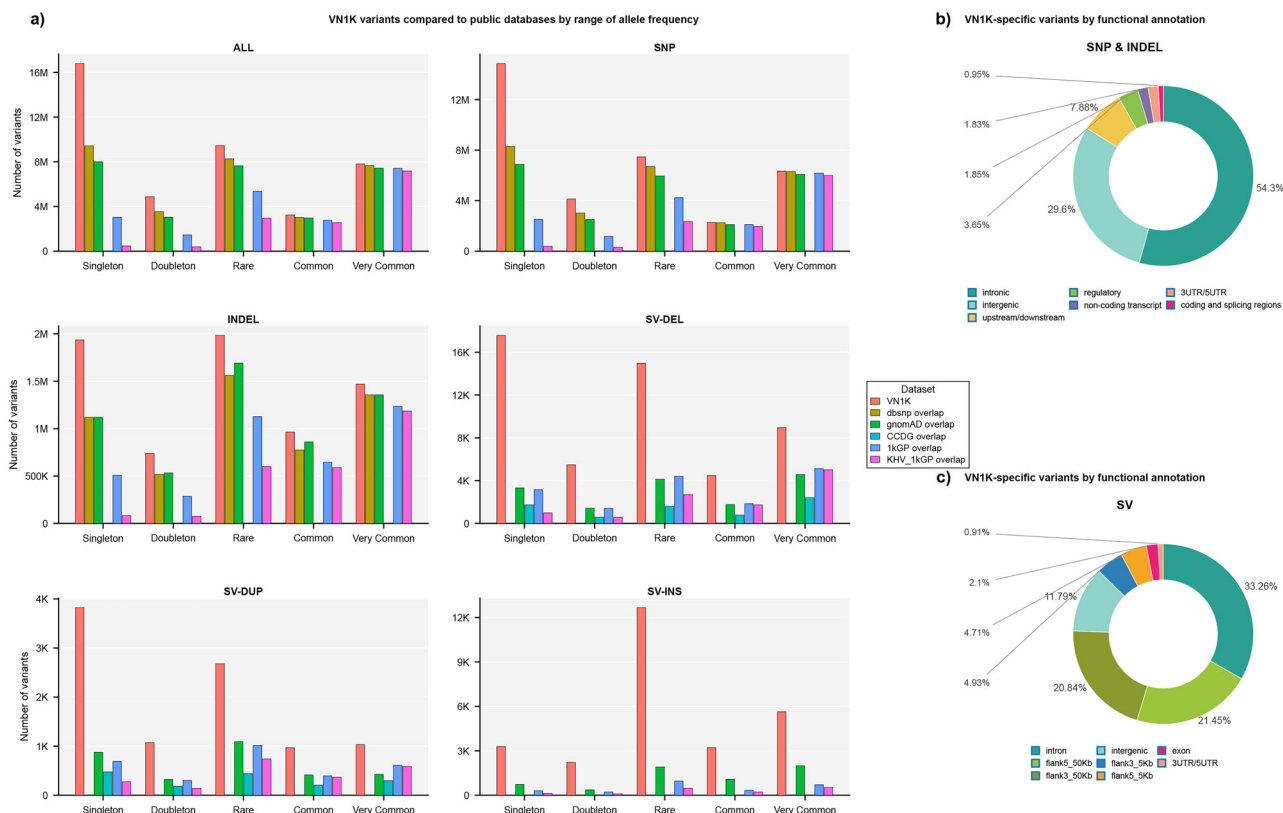


Fig. 3 | Functional analysis of the VN1K variants. **a** Distribution of SNPs, INDELs, and SVs in VN1K compared to public databases by range of allele frequency. Mutational profiles relative to public databases highlight rare mutations, doubletons, and singletons unique to the VN1K dataset. All variants were categorized by allele frequency (AF) and evaluated against the 1kGP-HC, gnomAD v3, and dbSNP151 datasets. SVs were categorized by AF and compared to the 1kGP-HC, gnomAD v3, and CCDG databases. **b**, **c** Summary of variant consequences based on functional

annotation of the VN1K-specific callset for SNPs, INDELs, and SVs. For SNPs and INDELs, 54.3% of variants were localized to intronic regions, 29.6% to intergenic regions, and 0.95% to coding regions. For SVs, 59.2% were intronic and 5.29% were exonic. Abbreviations: CCDG: Centers for Common Disease Genomics; AF: Allele Frequency; SNP: Single Nucleotide Polymorphism; INDEL: Insertion/Deletion; SV: Structural Variant.

observation that more than half of the Vietnamese carry the phenotype of intermediate metabolizer (IM). Two non-functional *CYP2C19*2* and *CYP2C19*3*, have AF of 26.37% and 5.21%, respectively. In contrast, *CYP2C19*17*, a more functional allele that is associated with ultra-rapid metabolism of drugs, has an AF of only 1.09%. Furthermore, we found that the 99 KHV samples in the 1kGP-HC dataset covered only 9472 variants assigned to 65 star-alleles, all of which were detected in VN1K, suggesting the comprehensiveness of our dataset in PGx gene regions. Specifically, top genes with highest Jost's D genetic distance between VN1K and 1kGP-HC (excluding KHV) were *VKORC1*, *CYP2D6*, *CYP2B6*, followed by *NAT2*, and *SLCO1B1* (see more details in Supplementary Fig. 17). More details of the VN1K pharmacogenes' AF can be found in Supplementary Data 6c. Taken together, our results present a comprehensive characterization of these important genomics regions, highlighting variants specific to Vietnamese.

Whole-genome methylation analyses reveal specific and functionally relevant epigenomic signatures

We posited that functional features of a reference genome are not only dependent on DNA sequences but also on the epigenomic states. We built a Vietnamese epigenomic reference starting with deep methylation profiling of two individuals (i.e., one male and one female) and 4 pools of 40 individuals using both PacBio and ONT technologies, followed by an integrative analysis of RNA-seq data²². We identified 27,605,743 highly confident methylation sites, representing approximately 95% and 98% of the sites from PacBio and ONT, respectively. The two results were consistent across all chromosomes, with Pearson

correlation coefficient (R) of 0.922 (Fig. 5a), although we noted that ONT data appeared to show more high-confidence sites (both methylated and unmethylated), whereas PacBio data showed more low-methylated and inter-methylated sites (Fig. 5c). Nevertheless, the consensus sites enabled us to investigate not only methylation patterns within pharmacogenomic regions for Vietnamese but also differentially methylated regions (DMRs) between Vietnamese and other ancestries of EUR and AFR, corroborating an additional layer of genomic diversity (Fig. 5b). We found DMRs within clinically relevant genes that are higher or lower in the Vietnamese samples compared to the European samples, as illustrated by the locus plot generated from Methyart²², where the similar score trends between the two sexes are found higher in the Vietnamese population (Fig. 5d). The comparison of expression of genes located in DMRs with the *GAPDH* gene as a control showed that the top differentially methylated genes, *DLGAP2*, also exhibited a lower gene expression level in the Vietnamese dataset compared to the European data, consistent with the results showing hypermethylation in Vietnamese and hypomethylation in European ($P < 0.15$ vs. $P < 9.22 \times 10^{-5}$) (Fig. 5e). In such locations, the SNP effect sizes generally estimated at the population scale in Caucasian would be different to that in Vietnamese due to different states of the genome, which may not be transcribed or accessible for transcription factor binding, even if a SNP presents. An analysis using RNA-Seq data for the genes associated with DMRs for 14 VN1K samples further confirmed the unique epigenetic signatures and consequences on gene expression. Many of those genes are enriched for important phenotypes such as alcohol drinks ($P < 6 \times 10^{-5}$), triglyceride metabolisms

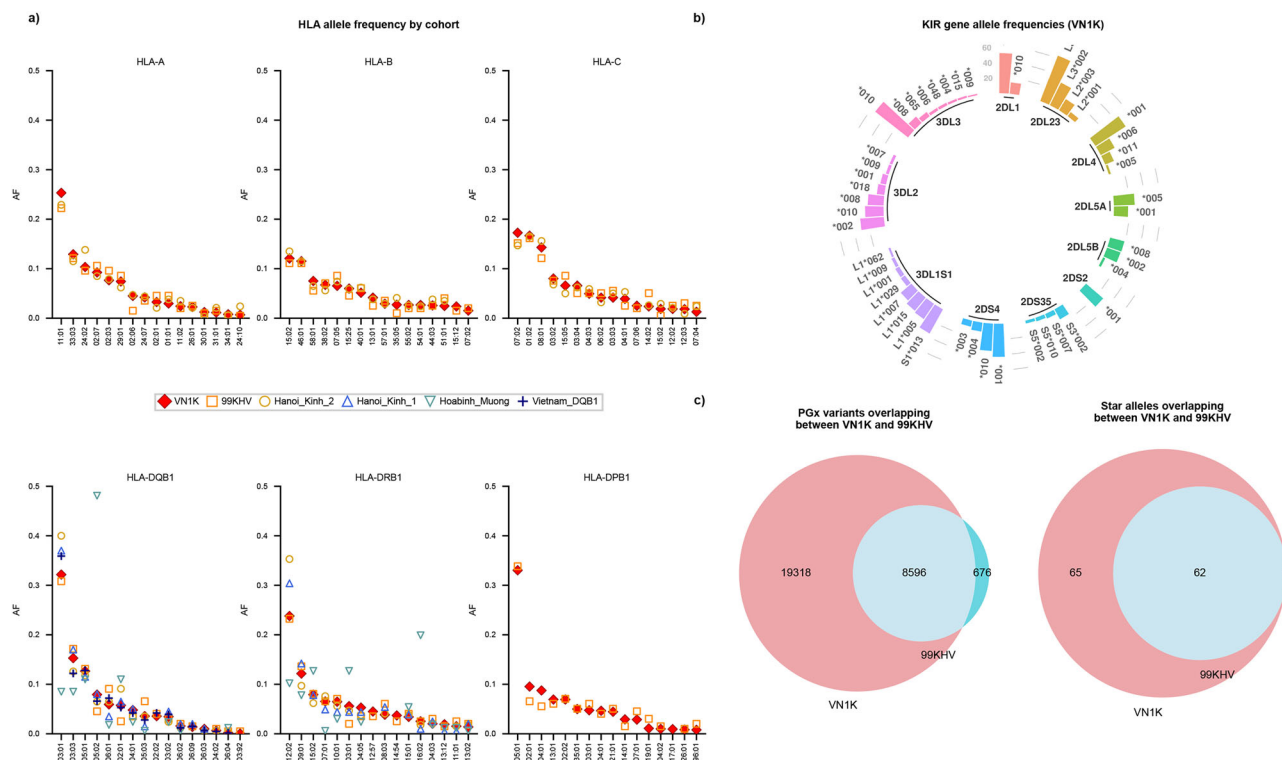


Fig. 4 | Immunogenomics and pharmacogenomics analysis. A comparative study involving VN1K and other existing datasets from Vietnam examined actionable genotypes and diplotypes. **a** The frequency of six *HLA* genes in the Vietnamese population from VN1K, KHV and public data from allele frequencies.net. Note that the Vietnam_DQB1 ($n = 2076$) obtained the result from genotyping data of gene

panel testing. **b** The AFs of KIR genes called from the VN1K dataset. **c** The number of PGx variants and star allele calls from the VN1K and the public dataset 1kGP-KHV (99 samples). Abbreviations: KHV: Kinh in Ho Chi Minh City, Vietnam (in the 1000 Genomes Project – 1kGP).

($P < 1e-3$), fatty lipid metabolism ($P < 2e-4$), and virus perturbation ($P < 2e-2$) (Fig. 5f). More results for this could be found in Supplementary Figs. 18–23. Taken together, such epigenetic and transcriptomic associated pathways suggest potential mechanisms for the observed higher incidence in liver-related diseases²³.

Multi-phenotypic traits and metagenomics analyses identify functional loci of interest

We further investigated the interplay between phenotypic traits and metagenomics of VN1K. First, an association analysis was performed with 12 quantitative clinical traits (Supplementary Data 7) and 8,536,859 variants from 810 samples with full trait information (Supplementary Data 1b) using additive genetic models. Despite the relatively small sample size, we found several independent loci surpassing the GWAS threshold ($P < 5.86e-9$, Bonferroni-corrected) (Fig. 6). These loci were significantly associated with six traits, including 1) triglyceride levels, 2) low-density lipoprotein cholesterol (LDL-C), 3) serum glucose levels, 4) *HbA1c*, and 5 and 6) the levels of two liver enzymes (i.e., alanine aminotransferase-ALT and aspartate aminotransferase-AST). These results were then further refined by mapping GWAS catalog data²⁴ to VN1K and comparing the VN1K common variants with those from other ethnic groups.

Second, we examined the presence and possible effects of common viruses in the genetic structure of the Vietnamese population. We found eight viruses with genetic material mapped to the VN1K genomes, in which hepatitis B virus (HBV) presented with the highest frequency (99 samples, or 9.8%), having mean read depth over 10 (Supplementary Fig. 24, Supplementary Table 3). These results provided a genetic-based assessment for the HBV epidemic in Vietnam, which is among the top 11 countries globally suffering from chronic viral hepatitis, with the prevalence of hepatitis B surface antigen

(HBsAg) over 8%²⁵. Such findings consistent between genetics data and epidemiology data warranted future validation studies, coupled with results from the epigenetic and transcriptomic analysis.

VN1K suggests refining genetic structure and evolution of the Vietnamese population

Vietnam was known to have a long and complex immigratory history with extensive ethnolinguistic and genetic diversity¹. This was examined more thoroughly in our analysis with ADMIXTURE²⁶ on the VN1K data combined with WGS data from the Human Genome Diversity Project (HGDP)²⁷ (Fig. 7, Supplementary Figs. 25–26). ADMIXTURE is a tool which uses multi-locus genotype data and applies a maximum likelihood framework to assign ancestry components to each individual. With $K = 10$ for ADMIXTURE analysis, our results showed that Vietnamese shared components with the Chinese Dai (CDX), Singapore-Malaysian (from SG10K), and Japanese (JPT). Notably, Group 1 and 2 (“group” referring to “genetic cluster”), major components for the Dai population (Chinese Dai in Xishuangbanna, China [CDX]) and the Vietnamese population (VN1K), are possibly specific to the East Asia Region. VN1K also aligned closely with CDX, consistent with previous reports^{28,29}. The genetic diversity was further highlighted by *Fst* values between Vietnam and other Asian groups in the HGDP. Our analysis showed a trend for the higher degrees of genetic difference of Kinh people for Central Asian groups than for East Asian groups (Supplementary Fig. 27). The notable difference in genetic distance as shown in our analysis suggests the need to include Vietnamese data in multi-ancestry analyses, especially among those ethnicities that have not been well studied.

To enable a systematic assessment of genetic architectures across Southeast and East Asian groups, we have performed cross-population analyses between VN1K and CPC dataset (Chinese Pangenome

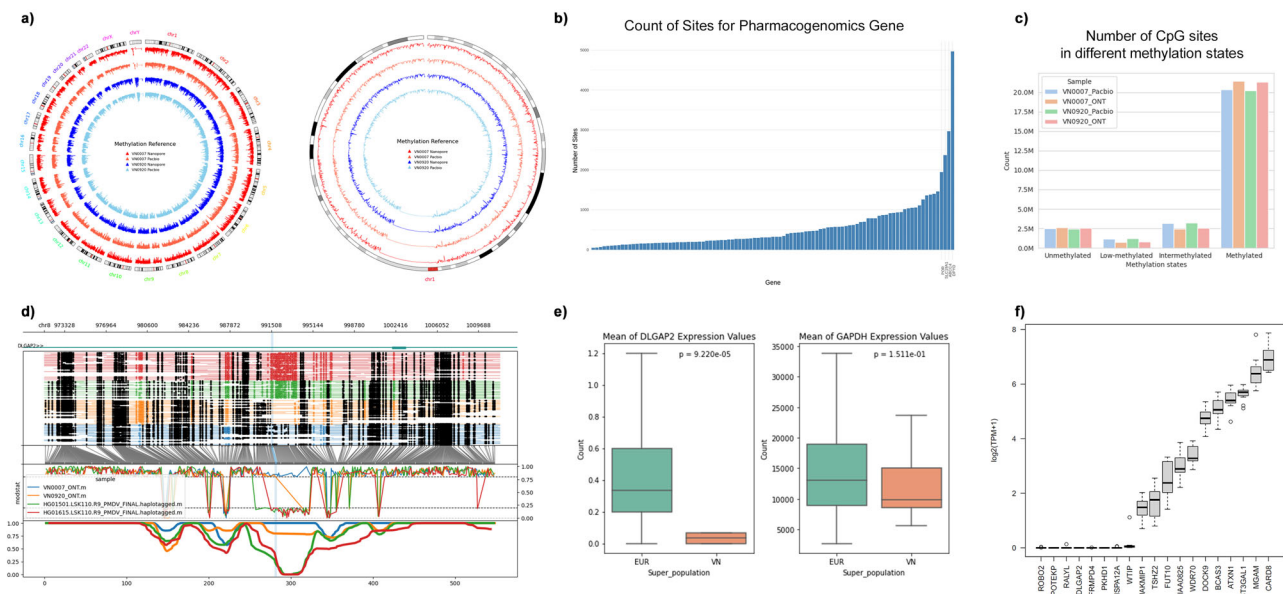


Fig. 5 | DNA methylation and gene expression analysis. **a** The pycirclize plot of the Vietnamese methylation reference panel for all chromosomes (1-22, X, Y) and the zoom in chromosome 1. **b** Consensus of CpG sites of Pacbio and ONT on 100 pharmacogenomic genes in PKSeq panel developed by RIKEN. **c** The distribution of methylation levels between Pacbio and ONT was categorized into four distinct groups. **d** The locus plot generated from Methylyst shows the differentially methylated regions (DMRs) between Vietnamese (VN0007 and VN0920) and European (HG01501 and HG01615) individuals were compared in gene *DLGAP2*. **e** The box plot compares the mean gene expression of genes located in differentially methylated regions (DMRs) to the expression of the *GAPDH* gene across European ($n = 481$) and Vietnamese ($n = 14$) populations. Each data point represents one

individual. Box plots display the median (center line), IQR (box bounds: 25th–75th percentile), and whiskers extending to $1.5 \times$ IQR; outliers are not shown. Statistical comparisons were performed using a two-sided Welch's t-test with no correction for multiple comparisons. *DLGAP2*: $t(15.42) = 5.24$, 95% CI [0.1930, 0.4569], $p = 9.20 \times 10^{-5}$. *GAPDH*: $t(14.86) = 1.51$, 95% CI [−1172.83, 6901.49], $p = 1.511 \times 10^{-1}$. **f** The methylation and RNA-seq gene expression of $\log_2(\text{TPM} + 1)$ are shown in the bottom panel of the figure (TPM: transcript per million). The selected gene set is enriched in alcohol drinking, lipids and other liver induced dataset. Box plots display the median (center line), IQR (box bounds: 25th–75th percentile), and whiskers extending to $1.5 \times$ IQR; outliers are shown as individual points.

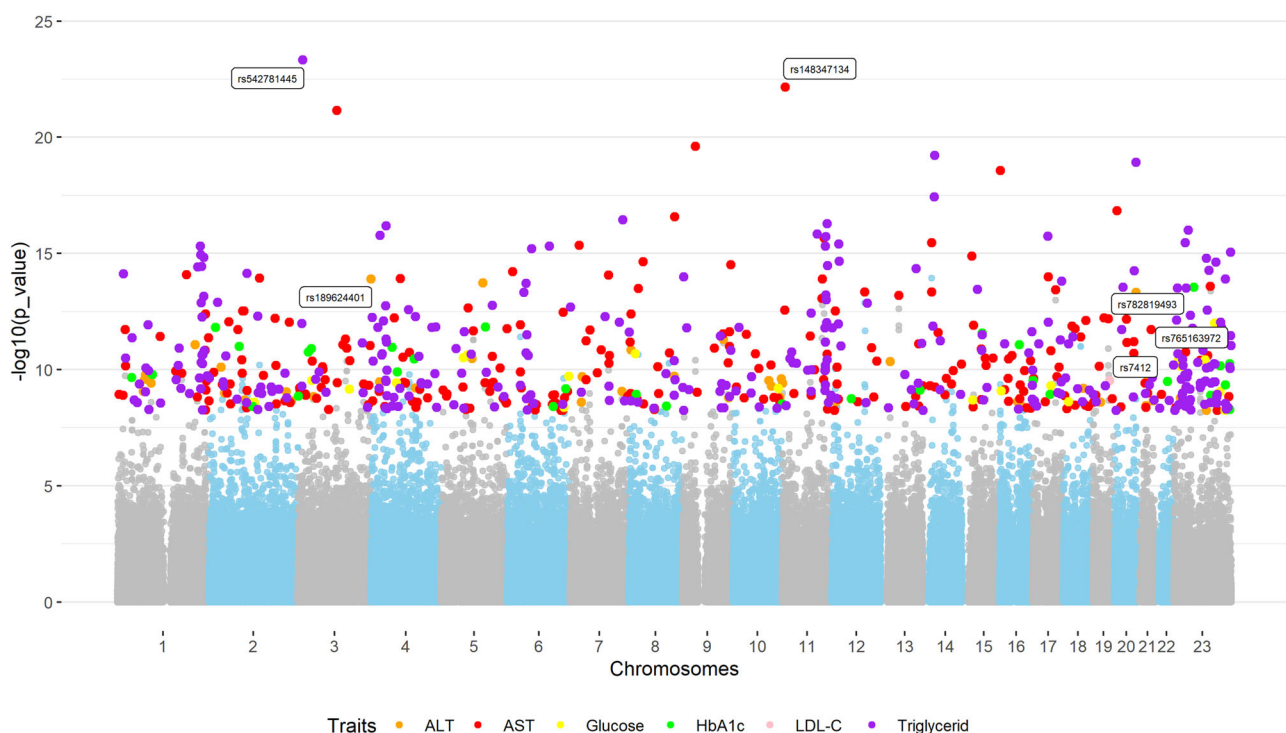
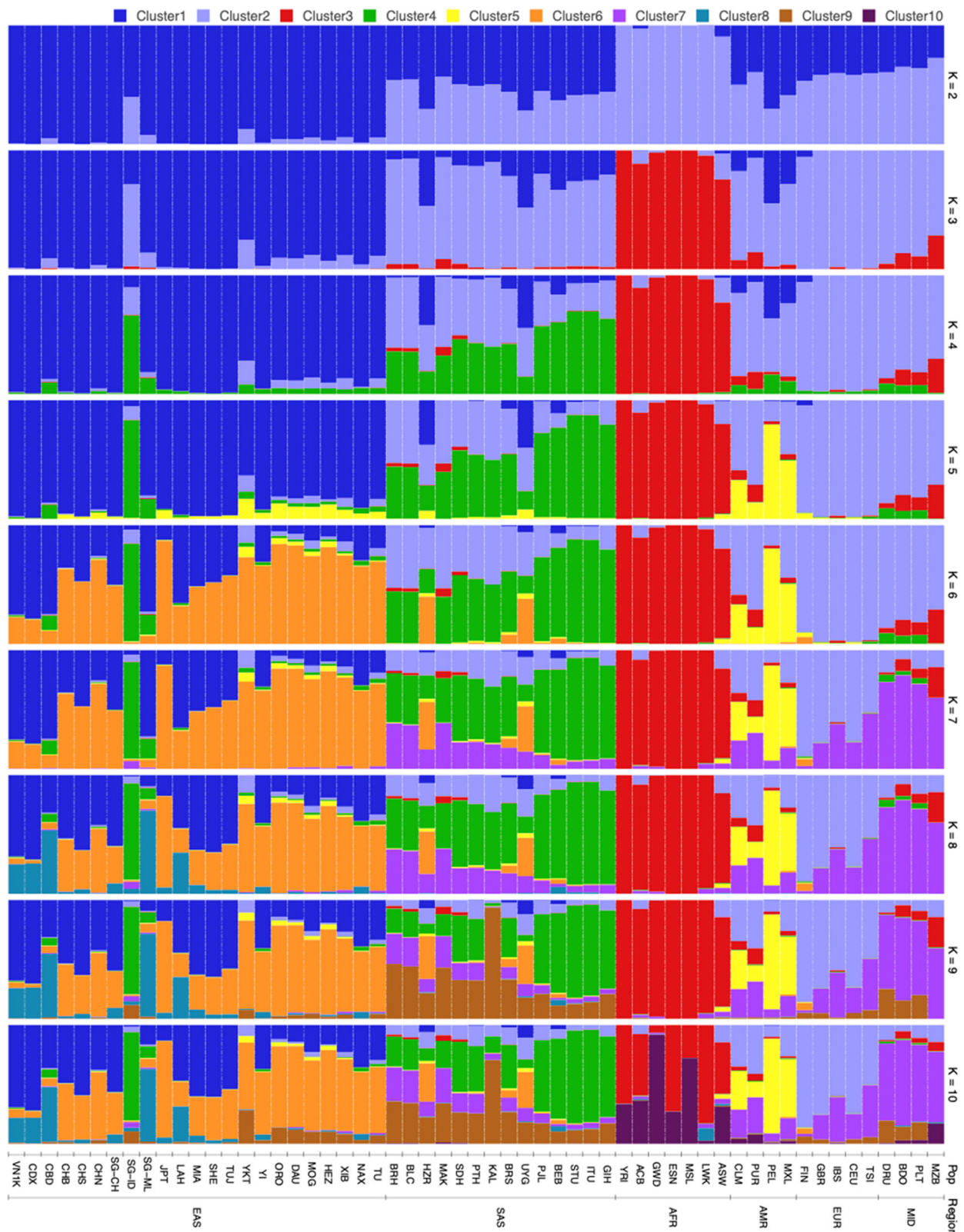


Fig. 6 | Multi-trait association analysis. 12 quantitative clinical traits and 8,536,859 variants from 810 samples in the VNIK dataset using additive genetic models. Manhattan plot revealed several independent loci surpassing the GWAS threshold

($P < 5.86 \times 10^{-9}$, Bonferroni-corrected significance). These loci were associated with six traits: triglyceride levels, LDL-C, serum glucose levels, Hemoglobin A1c, and levels of two liver enzymes (ALT/AST).



Consortium)³⁰. The CPC offered a high-quality and diverse long-read sequencing dataset, comprising 122 haplotypes from 61 samples across 36 East Asian populations. When comparing VNIK ($n=1011$) and CPC ($n=61$), the largest proportion of variants in both datasets is represented by singletons, as expected from population-scale sequencing. Rare variants ($AF < 0.01$) are almost exclusively detected in VNIK, compared with CPC, reflecting its larger sample size and greater power to

capture low-frequency variants. The overlap between the two is mainly concentrated in the common categories ($AF > 0.01$). Both VNIK and CPC also contain substantial numbers of their own common variants, highlighting population-specific genetic contributions. We next compared the SVs identified in VNIK and CPC. As the CPC comprises a small sample size, AF bins of SVs were defined using the VNIK dataset alone. With $AF \geq 1\%$, 14,650 of 26,045 (56.2%) SVs in VNIK were also present in

Fig. 7 | Population genetic structure analysis. The population structure signature of VN1K was shown based on the admixture ratio of ancestral components for each individual produced by ADMIXTURE. The results highlighted the genetic diversity within the target population by comparing values between Vietnamese and other Asian groups in the HGDP project. Abbreviations: VN1K (Kinh Vietnamese, our dataset), CDX (Chinese Dai in Xishuangbanna), CBD (Cambodian), CHB (Han Chinese in Beijing), CHS (Southern Han Chinese), CHN (Chinese), SG-CH (Singapore Chinese), SG-ID (Singapore Indian), SG-ML (Singapore Malay), JPT (Japanese in Tokyo), LAH (Lahu), MIA (Miao), SHE (She), TUJ (Tujia), YKT (Yakut), YI (Yi), ORO (Oroqen), DAU (Daur), MOG (Mongolian), HEZ (Hezhen), XIB (Xibo), NAX (Naxi), TU (Tu), BRH (Brahui), BLC (Balochi), HZR (Hazara), MAK (Makrani), SDH (Sindhi),

PTH (Pathan), KAL (Kalash), BRS (Burusho), UYG (Uyghur), PJJ (Punjabi from Lahore), BEB (Bengali from Bangladesh), STU (Sri Lankan Tamil from the UK), ITU (Indian Telugu from the UK), GIH (Gujarati Indian from Houston), YRI (Yoruba in Ibadan), ACB (African Caribbean in Barbados), GWD (Gambian in Western Divisions), ESN (Esan in Nigeria), MSL (Mende in Sierra Leone), LWK (Luhya in Webuye, Kenya), ASW (African Ancestry in Southwest US), CLM (Colombian in Medellin), PUR (Puerto Rican in Puerto Rico), PEL (Peruvian in Lima), MXL (Mexican in Los Angeles), FIN (Finnish in Finland), GBR (British in England and Scotland), IBS (Iberian populations in Spain), CEU (Utah residents with Northern and Western European ancestry), TSI (Toscani in Italy), DRU (Druze), BDO (Bedouin), PLT (Palestinian), MZB (Mozabite).

CPC. The extent of overlap was strongly dependent on SV type: INS showed extensive sharing, with 8383 of 10,041 (83.5%) overlapping overall, increasing from 76.0% in the 1–5% AF bin to 87.7% for AF > 5%. In contrast, DEL were much less shared, 6267 of 16,004 (39.2%) overall, from 27.0% to 45.4% across the same AF bins (Supplementary Fig. 28, Supplementary Table 4). The high concordance of INS can be interpreted that many of them are distributed across both Vietnamese and Chinese ethnics. Conversely, the lower DEL overlap likely reflects a combination of population-specific variants, driven by stronger natural selection and population structure maintaining DEL at lower rate, more Vietnamese specific frequencies, and technical discrepancies between short-read SV and assembly-based SV reconstruction.

VN1K data portal and multi-scale genome browser for integrated data resource

To enable the broad usage and impact of the current study, we have focused on making the VN1K data and results publicly available to the community through a platform named MASH (Management, Analysis, Sharing, and Harmonization) (<http://genome.vinbigdata.org>). MASH was designed as microservices with container-based infrastructure following the FAIR principle (Findable, Accessible, Interoperable, Reusable) to efficiently manage and analyze large and diverse data sources. MASH enabled searching gene names, rsID, variant ID, and other commonly genomic information. It also included a data harmonization module, displaying results for all common steps, including pre- and post-imputation quality controls, imputation, merging, and re-imputation. Importantly, MASH also integrated a genome browser which allowed researchers to work with both the linear and graph references, thus providing a dual architecture for visualizing and analyzing genomic variants and related information (Fig. 8). This integrated system not only facilitated accessing data but also optimized performance of analytic tools for diverse communities for the purposes of research, education and clinical application.

Discussions

International human genome projects, such as 1kGP and UK Biobank³¹, have shed light on understandings of human genetic diversity and their association with different phenotypes and diseases. Such resources provided an estimate of variant's AF in different populations³², which is critically important information to assess the likelihood and magnitude of causal variants of disease outcomes^{33–35} and drug responses^{36,37}. These databases also facilitate the imputation of genotypes missing in genome-wide association studies (GWAS) and increase the statistical power to discover associations between diseases and low-frequency variants^{38,39}. However, current studies have found that most genomic variants are rare and often occur at vastly different rates across populations, with many only appearing in specific populations^{2,40}. In addition, as the majority of genomic studies to date have been focused on Caucasian ancestry^{41–43}, millions of population-specific variants are missing. Therefore, genomic studies that capture local or specific population genetic makeups for specific racial and ethnic groups are essential.

Our study contributes to addressing the knowledge gap of human genetic diversity in specific racial/ethnic groups or populations, tackling the global issue in genomic equity and serving as robust genomic references not only for the 16th largest population, but also for related populations of Asian-origin. Although our sample size is still small compared to existing work in other countries, with 1,011 individuals representing 53 of the total 63 administrative divisions of Vietnam, VN1K remains a large-scale study with a comprehensive database of genomic variants for the Vietnamese population. We chose to study the Kinh ethnic group first for VN1K since they make up 86% of the entire population. Earlier studies have extensively explored the genetic architecture of Kinh people and other minorities using both single locus (Y-chromosome and mtDNA), genome-wide data and whole genomes^{1,44}.

Our study provided a graph-based pangenome reference named VPR for the Vietnamese population. VPR incorporated sequences from short- and long-read WGS, which helped reduce reference bias of European ancestry. As a result, VPR represented genetic diversity for the Vietnamese population, which helped enhance sensitivity of variant detection from SNPs to SVs. Our analytics also applied the latest methods to further capture the genetic diversity of Vietnamese, including deep learning approaches and ensembles of results from a comprehensive collection of robust methods for variant calling.

Although having a substantial role in human evolution and population diversity, study of SVs in underrepresented populations is scarce. Using cutting-edge sequencing technologies from PacBio and ONT platforms, we were able to produce a comprehensive reference for SVs, which are increasingly recognized to impact hugely on the phenotypes, especially disease traits. Latest deep learning tools, shown to better identify SVs, were applied. In this study, we assessed all possible SVs to achieve a consensus singular set including DEL, INS, DUP, and ITX. Such results enabled us to construct the profiling of various disease-related SVs and incorporate them into the VPR.

VN1K provided functional analyses on various important regions of the genome, in particular immunogenes, which has been long attracting great attention due to their critical roles in health conditions. Previous studies on the Vietnamese population have not used high quality, single-based resolution, whole genome analysis and were limited by small sample sizes. Using the high-depth whole genome sequencing data (30–60X), we could identify a large number (166) of HLA alleles. Besides, VN1K showed a high similarity in AF of most common alleles of both HLA-classes with previous studies, particularly the ones performed in Kinh population^{45–47}. Regarding KIR genomic variants, Amorim et al.⁴⁸ recently reported an investigation of KIR distribution in 140 unrelated Vietnamese healthy people from Hanoi, Vietnam using standard PCR-SSP technology, which is consistent with those observed from VN1K, again suggesting the comprehensiveness of our database. The additional information gained by our study compared to the previous studies might be attributable to sample size, the sensitivity and the specificity of genotyping methods and bioinformatic tools. For examples, the difference of variants called from

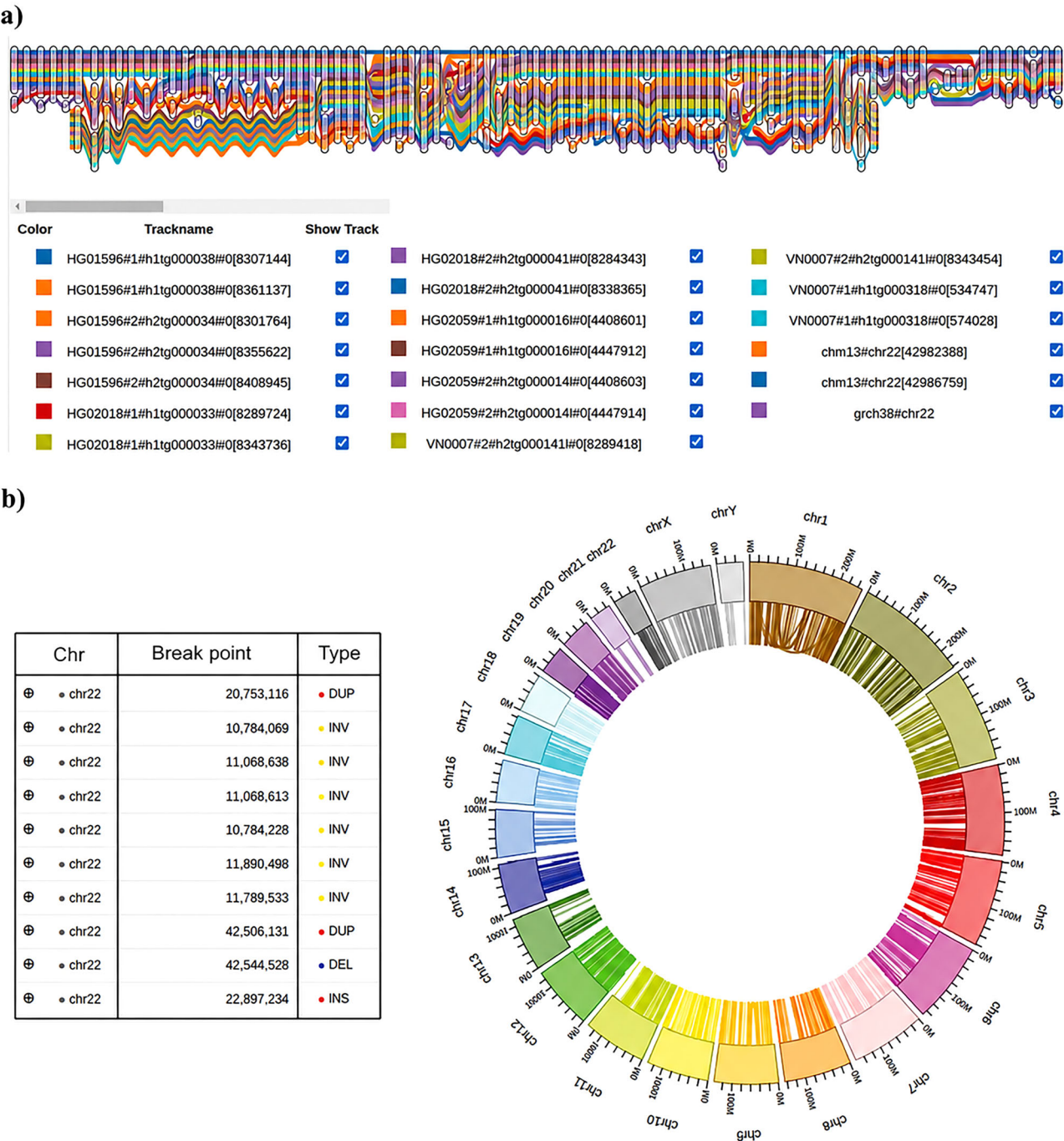


Fig. 8 | MASH data portal with an integrated genome browser. a The integrated genome browser, which allows researchers to explore all variants across chromosomes using both linear and graph-based genome references. The multiple paths of variants were shown by SequenceTubeMap as a genome graph. The colored lines

represent branches from genomic sequences of different persons. **b** The SV dataset was displayed on a variant table tab (left part) and visualized using pyCirclize software package (right part). The platform included a rich set of functionalities to explore SV distribution across chromosome, path graph, and gene annotation.

VNIK and 1kGP-HC for *2DL1* and *2DS1* suggesting further analysis on these genes to enable advancing immunogenomics research, potentially leading to improved clinical implications, including treatment and management strategies for various disease outcomes.

Due to resource limits, as is common in many low- and middle-income countries (LMIC), the primary limitation of this study is the relatively small sample size. Detecting rare variants therefore remains challenging with VNIK, although its sequencing depth exceeded typical standards (98% of samples >30x coverage, >80% of bases above Q30, median coverage >37x). Nevertheless, our analysis framework still yielded nearly 8.5 million variants that might be specific to

Vietnamese. We expect that the approach used in this work can be applied to other LMICs.

In summary, we characterized the genomic variations in the Vietnamese population and provided a comprehensive multi-omics and phenomics database together with a pangenome reference. This study could be used as a go-to resource for a wide-range of future genomic studies in Vietnam and beyond, especially in cross-ancestry analysis for populations of Asian-origin, which consist of 60% of the global population. This study is expected to produce impacts extending beyond research to helping with education, providing potential social and healthcare benefits for Vietnam and the world.

Methods

Ethics

The study design and conduct complied with all relevant regulations regarding the use of human study participants and received ethics committee approval from the Institutional Board Review (or Ethics Committee) of the Vinmec International Hospital, Hanoi, Vietnam (543/2019/QĐ-VMEC) and Hanoi Medical University, Hanoi, Vietnam (IRB-VN01.001/IRB00003121/FWA00004148, Approval No.: 362/GCN-HĐĐNCYSH-ĐHYHN). The study was conducted in accordance with the criteria set by the Declaration of Helsinki.

Sample collection

We collected peripheral blood samples of 1,108 healthy unrelated individuals self-declared as Kinh-ethnic for at least 3 generations. Kinh ethnic group was chosen because it is the largest ethnic group in Vietnam, accounting for 85.7% of the Vietnamese population (General Statistics Office of Vietnam's 2023 report). Unrelated relationships of participants were preliminarily checked by the interviewers and then cross-checked by genetic analysis on all samples. Sample collection was performed during 2019–2021 period at hospitals in Vinmec Healthcare System in four cities/provinces Ha Noi, Ha Tinh, Da Nang, and Ho Chi Minh City with participants originally come from 53 out of the total 63 cities/provinces across three main regions North, Central, and South in Vietnam. The distribution of samples in each region was roughly determined based on their population.

The sample selection was carried out using the health check examination provided by the Vinmec health-care service through the questionnaires. All participating individuals must be qualified by physicians and guided to sign a written consent form prior to joining the research. All participants received equal compensation for their involvement. The metadata of samples, which are de-identified, were collected to use for research purposes only. There are 34 traits collected in total, including demographic, anthropometric, biochemical, and clinical traits. Sex of participants was determined based on self-report and then cross-checked by genetic analysis to guarantee a balance between male and female samples.

DNA and RNA extraction, genotyping, sequencing and data quality control

Genomic DNA was extracted and the quantification was performed using Qubit dsDNA Broader Range Assay Kit (Life Technologies) or Quantifluor dsDNA System (Promega). DNA amount > 1 µg; DNA concentration > 35 ng/µL. Library preparation was done using Illumina protocol TruSeq DNA PCR-free High Throughput Library Prep Kit with IDT for Illumina TruSeq DNA UD Indexes (e.g., UD index was used to reduce index-hopping). All library clean-up steps were performed following the standard protocol of the manufacturer (i.e., Illumina). Quantitative PCR (qPCR) was carried out using the KAPA Library Quantification Kits for Illumina platforms (Kapa Biosystems) before sequencing. Subsequently, intact genomic DNA was fragmented and sequenced at Hi-tech Center of Vinmec International Hospital, Hanoi, Vietnam. Paired-end 150 bp whole genome sequencing with an insert size of 450 bp was performed on the Illumina Novaseq6000 platform. The target depth was 30x for all samples.

A subset of 96 samples from VN1K were randomly selected to be genotyped by the Axiom Precision Medicine Research Array (PMRA) at Novogene, Singapore. In addition, long-read sequencing was performed on another subset of our cohort on two platforms at the Institute for Molecular Bioscience, The University of Queensland, Brisbane, QLD, Australia. First, PacBio Sequel II platform was conducted in two (02) reference samples (a female and a male) and four (04) pooled samples (10 samples per pool, 40 samples in total) with 30X coverage. Next, Oxford Nanopore Technology (ONT) sequencing was performed for the aforementioned two reference samples with 30x coverage. These datasets not only enabled better detection of SVs

for the cohort but also provided DNA methylation information that could not be achieved with short-read sequencing data.

RNA extraction from blood samples were performed in GeneS-tory, Hanoi, Vietnam, then RNA samples were sent to Novogene, Singapore for sequencing with the following quality specifications: 1) Each sample must generate approximately 12 GB of FASTQ output; 2) The rRNA-depleted library must contain at least 40 million reads; 3) A minimum of 80% of bases must reach Q30 quality; 4) RNA integrity must be maintained with RIN values above 7; 5) rRNA and globin contents must each remain below 5%; 6) At least 80% of reads must map to the human genome; and 7) The GC content of total RNA must fall within the expected range of 39.3–55.7%.

For genomic data quality control, the raw fastq files were pre-processed by removing adapter sequences and trimming polyG in read tails which often happens in the Illumina two-color system. Furthermore, we used different filters, including removing reads with over 50% bases having quality or average base quality under Q30. PCA analysis will be performed for initial data quality control, and 10 samples that are genetic outliers outside the VN1K cluster were removed. The sample's sex was genetically confirmed, and two samples with contradictory declared and inferred sex were removed. Finally, we removed six samples that are predicted to be duplicate or having relative relationship with current samples. There are totally 1090 samples remaining after these filtering.

SNP and INDEL calling. The clean fastq files of each lane were then mapped against the reference genome GRCh38 p7 before being combined into one bam file. We used three state-of-the-art pipelines for variant calling: (1) Human_par pipeline which resembled bwa-0.7.12 and the GATK4 best practice pipeline with proper pseudo autosomal region ploidy values implemented on NVIDIA Parabricks (https://docs.nvidia.com/clara/parabricks/v3.5/text/human_par_pipeline.html), (2) Google DeepVariant on NVIDIA Parabricks v3.5, (3) Illumina Dragen v3.7 that was analyzed locally on the Illumina Dragen On-Premise Server with a hardware-accelerated FPGA card. To assure uniformity, we only kept the samples with the coverage at 15X that should be over 90% of the whole genome, and the improper pair percentage (IPP) be under 5%. We removed 79 samples that failed these filters. There are totally 1011 samples remaining after this step that are used in the final analysis.

For each pipeline, GVCF files were combined to obtain a single cohort VCF files using explicit tools recommended by each development team: 1) GATK GenomicsDBImport to combine the Human_par GVCF files, 2) Google GLnexus to combine the DeepVariant GVCF files, and 3) Dragen Genotyper to combine Dragen GVCF files. Next, we applied GATK VQSR to the call set of GATK GenomicsDBImport and the default hard filter of Dragen and DeepVariant pipelines. Only passing sites of each pipeline were selected and further processed by taking the consensus of at least 2 pipelines. More details of SNP and INDEL calling workflow could be found in Supplementary Fig. 29.

CNV calling. The Dragen CNV pipeline was used to call CNV per each sample and then merged all the CNV, based on the reciprocal overlapping of at least 80%. All CNV calls from Dragen were filtered to pass all of four conditions, including 1) CNV with length below 10000, 2) CNV with quality below 10, 3) CNV with low supporting number of bins with respect to event length, and 4) CNV with copy ratio within ± 0.2 of 1.0. The total CNVs passed all filters from 1011 samples is 628,301. The CNVs that are 80% or more overlapping were merged into one combined CNV. The start and end of combined calls are determined as the mean of start and end positions of individual CNVs. After merging, the combined list of CNVs consists of 20,241 CNV calls. To identify the VN1K-specific CNVs, we compared them with the structure variants of the Database of Genomic Variants released in 2020⁴⁹. We divided such loci into three categories, including 1) Common > 5%, 2) Intermediate from 1% to 5%, and 3) Rare > 1%.

Pangenome graph construction. To build the pangenome graph, we first implemented a comprehensive genome assembly workflow that integrated long-read data from PacBio using hifiasm v0.19.9-r616 (<https://github.com/chiyap123/hifiasm/releases/tag/0.19.9>) and ONT using Flye v2.9-gcc-10.3.0 (<https://github.com/mikolmogorov/Flye/releases/tag/2.9>) and polished by Medaka v1.12.0 (<https://github.com/nanoporetech/medaka/releases/tag/v1.12.0>). The resulting contigs went through quality assessment with QUAST v5.2.0⁵⁰ and then assembled into scaffolds guided by GRCh38, facilitated by ragtag v2.1.0⁵¹ to fill the remaining gaps and produce the final genome assembly. Then, the Vietnamese Pangenome Reference (VPR) was built by comprising core assemblies and incorporates variants derived from both long-read and short-read sequencing data of VNIK. For the core assemblies, the PanGenome Graph Builder (PGGB) v0.6.0⁵² was used to create a graphical fragment assembly (GFA). This tool processed multiple genome assemblies as input, including the GRCh38 reference, the recently released T2T-CHM13 assembly¹⁴, two KHV assemblies from HGSVC dataset⁵³ (HG02018 and HG01596), and two VNIK assemblies (VN0007 and VN0920), which were produced using a hybrid approach combining PacBio and ONT technologies. Subsequently, vg v1.56 (<https://github.com/vgteam/vg/releases/tag/v1.56.0>) was employed to integrate into the VPR the genomic variants (SNPs, INDELs, and SVs) identified from VNIK, including PacBio long-read sequencing. By aligning the genomic data of two KHV samples, HG02080 (HPRC) and HG02059 (HGSVC), against the VPR, we could identify unknown sequences and SVs and seamlessly integrate them into the expanding pangenome graph. More details of pangenome graph construction could be found in Supplementary Fig. 30.

SV calling. The workflow for SV calling included four steps: First, SV calling was processed using pbsv 2.9.0 for PacBio data and CuteSV 2.0.2⁵⁴ for ONT data and then filtered by bcftools 1.13. In detail, we retained only the chromosomes within the range of [1-22XY], while discarding others. After that, quality control was performed for all structural variants (SV) with criteria involving filtering, read depth, length, and quality score of SV. Second, a deep learning model named cnnLSV⁵⁵ was used to convert each SV into an image and predict SV types. The training process for the cnnLSV model included two stages: filtering and detecting. In the filtering stage, existing callers were used to refine detection results, labeling true and false positive SVs, and then converting them to images. Next stage, the images were applied to train models to detect SVs. Subsequently, we employed the SURVIVOR 1.0.7⁵⁶ to merge the SVs which consolidate overlapping or conflicting calls, setting a maximum measured pairwise between breakpoints limit of 1000 and SV length at least 50 bp. This workflow ensured the quality of the SV by incorporating an image-based approach for better detection of SVs. More details of the SV calling process could be found in Supplementary Fig. 31. Parameters for running tools in this part could be found in Supplementary Methods.

Reference panel construction. We have constructed a haplotype reference panel for phasing and imputation based on the database of genomic variants created by VNIK. From the final dataset, the singleton variants were removed using VCFtools version 0.1.16⁵⁷. We filtered out multi-allelic sites via the BCFtools norm version 1.9⁵⁸. SHAPEIT4⁵⁹ was used to phase the haplotype of all samples with default settings. Finally, Minimac4⁶⁰ was used to impute phased variants with our haplotype reference panel. For comprehensive analysis and comparisons, we exploited five imputation panels, including 1kGP, SG10K, VNIK, and two panels that combined VNIK with 1kGP and SG10K, respectively, which were processed in the same way as the VNIK reference panel. Two combined panels were respectively merged with the VNIK dataset following guidance from Huang et al.⁶¹. Parameters for running tools in this part could be found in Supplementary Methods.

Imputation evaluation. A total of 96 unrelated individuals were randomly selected from VNIK and genotyped with Axiom Precision Medicine Research Array (APMRA) (Applied Biosystems) to evaluate the VNIK imputation reference panel. The remainder of the VNIK dataset, comprising 915 samples (named VN915) was used as a reference panel. The imputation pipeline was performed on chromosome 20 using Minimac4 with five imputation panels, including three individual panels (VN915, 1kGP-HC, SG10K) and two merged panels (VN915_1kGP-HC, VN915_SG10K). The genotyping data was then imputed and compared to the WGS data to measure the aggregate R2 (Pearson correlation) and NR-allele concordance rate. We computed the aggregated R2 between imputed and true genotypes (determined based on WGS data). The number of high-quality variants was measured with Rsq value, which determined how well the variants were imputed from the reference panel (assuming Hardy-Weinberg equilibrium). We counted the number of variants having Rsq ≥ 0.8 (high-quality variants) in each MAF group for comparison purposes specifically for rare and low-frequency variants, which were frequently challenging to impute effectively. We evaluated the imputation error rate on 24 Asian populations from HGDP data, which contained the genotyping data generated with Illumina HuHap 650k microarray. Parameters for running tools in this part could be found in Supplementary Methods.

HLA imputation. We have developed a model for HLA imputation which employs a principal component analysis (PCA) algorithm with the aim of retaining the variants that carry the most valuable information and reduce the input dimensions at the same time. The problem of linkage disequilibrium was handled through the ability of convolutional neural networks (CNN), which represented data with different layers, and efficiently aggregate information from neighbor variants through the sliding window. We built a separate model for each group of HLA genes according to their position on the gene. The HLA alleles were called using 3 different methods and consented through majority voting^{62–64}. In order to preserve diploidy information and ignore position information, we encoded them with 2 operations, AND and OR. The AND operation indicated whether both haploids have variation at this location, while the OR operation indicated whether one variation on both haploids. With this encoding, we eliminated the workload of generalizing HLA information from the exact haplotypes and made the model work on the CNN most effectively. The Binary Cross Entropy loss function was used and the HLA variants were randomly divided with a ratio of 9-1 to obtain the training and validation datasets.

Functional genomic, immunogenomic, and pharmacogenomic analysis. Ensembl VEP⁶⁵ was used to annotate the variants to predict their effect for functional analysis. Mobile Element Insertion (MEI) analysis was done on both short- and long-read sequencing data using xTea^{66,67}. Parameters for running tools for genomics annotation and MEI analysis were described in Supplementary Methods.

For HLA analysis, WGS-derived HLA sequence data of each individual in VNIK and 1kGP-HC were imported into xHLA for both HLA class I and class II genotyping with resolution of four digits. Fisher's exact test was used to assess the differences in allele frequencies between the Vietnamese (VNIK) and global populations (1kGP-HC) with $p < 0.05$ to be the threshold for a significant difference. HLA typing was then performed from paired-end WGS data of VNIK and 1kGP-KHV (99 samples) using HISAT-genotype (version 1.3.2, built on HISAT2 2.2.1). Raw sequencing reads from both forward (R1) and reverse (R2) strands were used as input for allele assignment. The analysis was conducted using default parameters as implemented in HISAT-genotype. HLA alleles were inferred at high-resolution (four-field) level for classical loci including *A*, *B*, *C*, *DRB1*, *DQA1*, *DQB1*, *DPB1*, *DPA1*. Parameters for running tools for HLA analysis were described in Supplementary Methods.

The genotype of the KIR complex was called by two tools, KPI⁶⁸ and KIRCLE⁶⁹. KPI systematically investigated gene-specific small markers (*k-mer*) of KIR genes and then applied those markers to a synthetic probe algorithm to identify the presence/absence of 16 KIR genes and 16 haplotype structures. KIRCLE, on the other hand, was used to infer alleles and genotype for each KIR gene from aligned WES data in BAM/CRAM files. KIRCLE was performed through four major steps: 1) pre-processing, 2) local alignment with BLAST, 3) bootstrapped expectation-maximization, and 4) thresholding. After obtaining KIR variants, Fisher's exact test was employed to compare the AFs across all these variants.

Pharmacogenomic phenotype prediction was performed using PyPGx (version 0.26.0)⁷⁰ on BAM files derived from 1011 VN1K samples. The PyPGx NGS pipeline processed aligned sequencing data through multiple analytical steps, including variant extraction from VCF, copy number estimation, haplotype phasing, and diplotype assignment. Overview, the pipeline assigned star alleles (e.g., *CYP2D6**1/*10, *CYP2C19**2/*2, *NUDT15**1/*3) and translated them into predicted phenotypes (e.g., normal/intermediate/poor metabolizer) according to established pharmacogenomic guidelines. The final output was provided in tabular format, including key fields such as sample ID, gene, diplotype, predicted phenotype, and supporting variant annotations.

Furthermore, Jost'D distance was used to calculate the distance between the VN1K and other populations based on pharmacogenes with the following equation:

$$D = \frac{H_T - H_S}{H_T}$$

where:

- H_T is the total heterozygosity (genetic diversity) in the combined population.
- H_S is the average heterozygosity within subpopulations.

Methylation and gene expression analysis. A DNA methylation reference panel was constructed based on ONT and PacBio data from two VN1K samples (VN0007 and VN0920, one male and one female). ONT data was generated from R10.4.1 chemistry that outperformed R9.4.1 in methylation calling, while PacBio data utilized HiFi reads, following the respective manufacturer's best practices for DNA methylation calling, as detailed in Supplementary Fig. 32. In the final step, methylation scores were called from 5mC BAM files based on the MM and ML tags. To standardize the final format of the two, we used Modkit to generate the final BEDGraph files for both technologies. Subsequently, for each sex, we created a consensus by overlapping sites with common positions based on chromosome, start, and end coordinates. The final DNA methylation reference panel was constructed by overlapping the consensus files of the two sexes, with chrY taken directly from the male consensus file. Next, we compared the Vietnamese methylation reference to other populations to identify differentially methylated regions (DMRs) between them. Due to the differences in technologies between the VN1K and the 1kGP methylation datasets, we corrected batch effects using the limma package⁷¹. Methylation sites were categorized as hypermethylated and hypomethylated based on their methylation levels (>0.75 for hypermethylation, <0.25 for hypomethylation). Sites that were consistently hypermethylated in the Vietnamese population but hypomethylated in at least one other population of African (AFR), Admixed American (AMR), East Asian (EAS), European (EUR), South Asian (SAS) or vice versa were retained for further analysis. Enrichment analysis was then performed using Fisher's exact test in the LOLA package⁷², and gene annotation was carried out with the annotatr package⁷³ to explore patterns within the DMRs.

To analyze expression levels of the DMR-associated genes, we exploited RNA-Seq data from 14 samples and compared them to the

gene expression profiles of the 1kGP samples. Raw paired-end RNA-seq reads were subjected to quality control using fastp (v0.23.1) (<https://github.com/opengene/fastp>) to remove adapter sequences and low-quality bases. Cleaned reads were quantified using Salmon (v1.5.2) (<https://github.com/COMBINE-lab/salmon>) in quasi-mapping mode against the reference transcriptome. Transcript-level abundance estimates were then imported into R using tximport (v1.26.1) (<https://bioconductor.org/packages/release/bioc/html/tximport.html>), and aggregated to obtain gene-level raw counts. Gene symbols were then annotated using the org.Hs.eg.db package (v3.16.0) (<https://bioconductor.org/packages/release/data/annotation/html/org.Hs.eg.db.html>). To correct for potential batch effects between Vietnamese (VN) and European (EUR) samples, we applied ComBat-seq from the sva package (v3.46.0) (<https://www.bioconductor.org/packages/release/bioc/html/sva.html>). The batch-adjusted expression matrix was subsequently used for comparative analyses between populations to evaluate whether genes located within DMRs exhibit significant expression differences. Parameters for running tools for RNA-Seq data analysis were described in Supplementary Methods.

GWAS analysis. A genome-wide association study was performed with 810 samples with a full lab test for 12 traits. Under an additive genetic model, GWA analysis was carried out using linear regression using best practices (e.g., linear regression was used to test the association between genomic variants and quantitative traits collected in the study). The covariates that we included were year of birth, gender, weight, height and the first 10 principal components. PCA were then performed from post-processed variants that were pruned via PLINK v1.9. The results were further filtered with the threshold of 5.86×10^{-9} . The final set of loci were investigated with Ensembl VEP⁶⁵ and GWAS catalog (ver. 2022-12-21)²⁴.

Population genetic structure analyses. We used the liftOver⁷⁴ program to convert the coordinates of the SNPs of the SG10K data from hg37 to hg38. After this, there were reference alleles in the SG10K that must be changed to alternate alleles in hg38 coordinates, and vice versa. The liftovered SG10K dataset included 4,810 individuals belonging to three main ethnic groups (Indian, Chinese, and Malaysian). All SNPs shared among VN1K, 1kGP, HGDP, and SG10K were merged into a single VCF file using bcftools⁵⁸. The merged dataset contained 9254 individuals belonging to 108 populations and 1,627,296 overlapping SNPs. Because the large number of samples in SG10K was overrepresented by only three ethnicities, we randomly selected 200 samples for each population in SG10K for downstream analyses⁷⁵. In addition, we removed populations in the HGDP dataset that had fewer than five samples. The final dataset used for population structure analyses consisted of 4,626 individuals belonging to 57 populations.

Before ADMIXTURE analysis was performed on the above 57 populations, we filtered out SNPs with $MAF \leq 0.05$ or the missing rate ≥ 0.05 then pruned SNPs with high correlations using indep-pairwise in PLINK⁷⁶. We performed PCA analysis by PLINK on the VN1K dataset after pruning variants with high linkage disequilibrium and $MAF \leq 0.01$. We then split samples by their gender and original region to visualize different principal components. The final genotypes of 57 populations included 130,120 independent SNPs in PLINK binary format. Then, at each K from 2 to 35, the files were analyzed by the ADMIXTURE 10 different seed values with 5-fold cross-validation. We chose the best optimal cross-validation error at each K and took an average of each cluster in each population after using CLUMPAK⁷⁷ to analyze results from different seeds. Finally, we visualized the posteriors by the pophelper library⁷⁸.

For VN1K-CPC genomic comparative analysis, we used the standard version of the CPC dataset with 18,290,517 million variants across chromosomes. CHM13v2-based variant callset was converted to GRCh38-based one using LiftoverVcf from Picard v2.20.4. According

to the CPC website, variant files reflected different levels of filtering: “-full” files retain all chromosome-assigned contigs without masking or clipping; “-min.af.0.1” files were optimized for short-read mapping by preserving nodes covered by at least 10% of haplotypes.

Data portal construction. We have developed a platform called MASH (Management, Analysis, Sharing, and Harmonization) with optimized modules for storing and sharing data, and providing a user-interface to explore all sample and variant information in VN1K (and subsequent datasets). MASH was developed based on Gen3, a well-known platform from The University of Chicago (<https://gen3.org>), which was originally developed for NCI Genomic Data Commons (<https://gdc.cancer.gov>) to manage large genomic datasets. MASH provided a workflow to accelerate the process of genomics data analysis, which included well-known tools such as BWA, GATK, Google DeepVariant, Ensemble VEP, and in-house tools and pipelines^{79,80}. MASH also included a genome browser which integrated both linear-based references (such as GRCh37/GRCh38) and graph-based references (such as VPR). Leveraging MoMI-G⁸¹ as the underlying framework, the Genome Graph Browser offered a user-friendly interface to visualize SVs by a variation graph. The browser boasted a rich set of functionalities, including functionalities to explore SV distribution across chromosome, path graph, and gene annotation. To achieve these visualizations, SVs were called to compute into a dedicated dataset. This dataset was then displayed on a variant table tab and visualized using pyCircize 0.10.1 version⁸². Furthermore, SequenceTubeMap was assembled into the genome graph browser for SV graph visualization. For input, the browser accepted variation graphs in XG format, along with the metadata associated with the SVs. Additionally, it could be processing graph alignment information (GAM) derived from BAM files. All these inputs were conveniently specified within a configuration file using YAML format. Gene annotation function was designed to retrieve from Ensembl or regions listed in BigBed format. In summary, the user interface was built using TypeScript and implemented for testing on Linux operating systems.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

VN1K multi-omics and phenomic data generated in this study are available in MASH Data Portal [<https://genome.vinbigdata.org>], which was deployed on on-premises infrastructure to facilitate data storage and backup. Population-level data are publicly available, except information such as allele frequency, with clear Terms of Use, Privacy Policy, and other related statements on the MASH. The individual-level data are available under restricted access to protect participant privacy and comply with Vietnamese laws and regulations governing cross-border data transfers, except information such as Ethnicity, Sex, Sample Collection Area, and Original Area. The restricted data can be obtained by ordering through the MASH, which requires registration of an account, selecting the data items, and submitting a research proposal to get approval. Data access is also available through collaborative research with the VN1K Project, subject to discussion with the corresponding author Nam S. Vo (nam.vs@vinuni.edu.vn). The requests are granted based on evaluation of the research proposal by the VN1K Data Access Committee with permission/notification from/to Vietnam's Government. Resolutions regarding access requests typically take between 3 and 4 weeks. After this, the data can be downloaded through the portal within 1–2 months. For further data access requests such as processed data for specific purposes please contact the above main corresponding author. This study also uses the following public datasets: the genome reference GRCh38 p7 and the indexed files were downloaded from the resource bundle on the GATK website (<https://console.cloud.google.com/storage/>

[browser/genomics-public-data/resources/broad/hg38/v0/](https://browser.genomics-public-data/resources/broad/hg38/v0/)); the T2T-CHM13 assembly dataset was downloaded from the NCBI data portal (https://www.ncbi.nlm.nih.gov/datasets/genome/GCF_009914755.1/); the IGS resource can be used to access the 1kGP variant datasets [https://www.internationalgenome.org/data-portal/data-collection/1000genomes_30x] and the HGDP variant datasets [<https://www.internationalgenome.org/data-portal/data-collection/HGDP>]; the KHV assemblies are publicly available through the IGS resource with the HPRC (phase 2) [<https://www.internationalgenome.org/data-portal/data-collection/hprc2>] dataset and the HGSVC (phase 3) [<https://www.internationalgenome.org/data-portal/data-collection/hgsvc3>] dataset; the variant dataset for gnomAD version 3 are available on gnomAD [<https://gnomad.broadinstitute.org/downloads/#v3-variants>]; the dbSNP variant dataset version 15 is publicly available through the NCBI data portal [https://ftp.ncbi.nih.gov/snp/organisms/human_9606_b151_GRCh38p7/]; the CCDG variant dataset is available through the CCDG on <https://www.ccdg.rutgers.edu/mapping-and-characterization-structural-variation-17795-human-genomes> and https://github.com/fakedrtom/SVAFotate/tree/master/supporting_data; the SG10K dataset was accessed through the EGA portal with the SK10K Pilot approved study “Building databases of genomic variants for Vietnamese population” [https://www.a-star.edu.sg/docs/librariesprovider11/npm/approved-sg10k_pilot_projects_02apr26.pdf?sfvrsn=31eb18e4_1], under accession number EGAD00001005337; the CPC assemblies and variants dataset are available on the NCNGC portal [<https://ngdc.cncb.ac.cn>] under BioProject ID: PRJCA011422, and are also available on the CPC portal [<https://pog.fudan.edu.cn/cpc/#/data>]. The GWAS association dataset (v1) is available on the GWAS Catalog data portal [<https://www.ebi.ac.uk/gwas/api/search/downloads/associations/v1.0>]. Source data are provided with this paper.

Code availability

VN1K source code is publicly available on GitHub [<https://github.com/VinGenome/VN1K>]. The source code is also publicly available on Zenodo [<https://doi.org/10.5281/zenodo.19677211>].

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Author contributions

N.S.V. conceived, conceptualized, designed, and supervised the whole project. V.H.V. and L.L. conceived and supervised part of the project. D.H.L. and Q.N. conceptualized, designed, and supervised part of the project. V.S.L., N.T.D., M.D., T.T., and L.T.N. supervised part of the project. T.T.H.T. and T.H.H. conducted the primary data analysis and interpretation. N.T.N., D.T.N., T.M.P., N.N.N., G.M.V., V.C.D., Q.T.V., T.K.N., S.V.N., H.Q.V., T.M.N., D.T.N., H.T.T.N., N.Q.L., Q.H.N., C.T.H., and Q.T. performed the data analysis. M.H.T., T.P., L.T.L., D.X.D., and G.H.P. performed the data curation. M.H.T., D.M.V., H.T.T.L., T.D.N., L.T.N., Y.H., D.X.D., and G.H.P. performed the sample collection. D.M.V., H.T.T.L., and L.N. carried out the sequencing experiments. T.D., H.N., T.D., and C.L. built the data portal. T.T.H.T., T.H.H., N.T.N., T.M.P., N.N.N., Q.T.V., G.M.V., V.C.D., T.K.N., S.V.N., Q.N., and N.S.V. drafted the manuscript. N.Q.L., H.T.T.N., D.M.V., T.D.N., H.N.L., V.S.L., N.T.D., D.H.L., and Q.N. revised the manuscript. N.S.V. revised and wrapped the final manuscript. All authors have read and approved the final manuscript.

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Competing interests

T.M.P., N.N.N., V.C.D., Q.T.V., and N.S.V. are current employees of GeneStory JSC, Vietnam, a company that develops and markets products for genetic testing. The other authors declare no competing interests.

Additional information

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